

MAT11-5

Descriptive Statistics & Data Reasoning

Summarizing, visualizing, and critically interpreting data

Mathematically disciplined interpretation, not dashboard production.

Albert School — 22 hours

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1 Central tendency

For a dataset $x_1, \dots, x_n$:

- **Mean:**

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

- **Median:** the middle value when the data are sorted (the average of the two middle values when n is even).
- **Mode:** the most frequent value.

Choosing the appropriate measure depends on the distribution shape and the presence of outliers:

- The mean uses every value and is appropriate for roughly symmetric data without extreme outliers.
- The median is unchanged by extreme values and is appropriate for skewed data or data with outliers (for example incomes, house prices).
- The mode is appropriate for categorical data or to report the most common value.

On right-skewed data the mean exceeds the median; on left-skewed data the mean is below the median; on symmetric data they coincide.

2 Dispersion

For a dataset $x_1, \dots, x_n$ with mean $\bar{x}$:

- **Variance:**

$$s^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2.$$

- **Standard deviation:** $s = \sqrt{s^2}$, in the same units as the data.
- **Interquartile range (IQR):** $IQR = Q_3 - Q_1$, where Q_1 and Q_3 are the first and third quartiles (the 25th and 75th percentiles).

The standard deviation measures spread around the mean and is sensitive to outliers; the IQR measures the spread of the middle 50% and is robust to outliers. Dispersion is interpreted relative to the data-generating process: the same standard deviation means high relative variability for a tightly controlled process and low relative variability for a naturally variable one.

A **box plot** displays the five-number summary — minimum, Q_1 , median, Q_3 , maximum — as a box from Q_1 to Q_3 with a line at the median and whiskers to the extremes. Points beyond $Q_1 - 1.5 \times \text{IQR}$ or $Q_3 + 1.5 \times \text{IQR}$ are drawn separately as outliers.

3 Bivariate statistics

For paired data $(x_1, y_1), \dots, (x_n, y_n)$ with means $\bar{x}$ and $\bar{y}$:

- **Covariance:**

$$\text{cov}(x, y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}).$$

- **Pearson correlation:**

$$r = \frac{\text{cov}(x, y)}{s_x s_y}, \quad -1 \leq r \leq 1,$$

where s_x and s_y are the standard deviations.

The sign of the covariance and of r gives the direction of co-movement: positive when the variables tend to rise together, negative when one rises as the other falls. The magnitude of r gives the strength of the **linear** relationship: $r = \pm 1$ is a perfect line, $r = 0$ is no linear relationship. Unlike covariance, r is dimensionless and unaffected by the units of x and y .

A **scatter plot** shows each pair (x_i, y_i) as a point and reveals the form, direction, and strength of the relationship.

Common pitfall **Correlation does not imply causation.** A non-zero correlation between two variables can arise without either causing the other — through a common cause (a confounding variable), a coincidence, or reverse causation. Example: ice cream sales and drowning deaths are positively correlated, but the common cause is hot weather, not one causing the other.

4 Statistical biases

A bias is a systematic distortion that pushes conclusions away from the truth.

- **Sampling bias:** the sample is not representative of the population. Example: a political poll conducted only by landline telephone overrepresents older voters.
- **Selection bias:** the process that includes units in the data is related to the outcome studied. Example: evaluating a treatment only on patients who completed it ignores those who dropped out because it was not working.
- **Survivorship bias:** only the units that “survived” a process are observed. Example: analysing returning World War II aircraft for armour placement misses the planes that were shot down — armour was needed where the survivors were **not** hit.
- **Confirmation bias:** evidence is sought or interpreted to confirm a prior belief. Example: an analyst who expects a product to succeed notices positive reviews and dismisses negative ones.

4.1 Simpson’s paradox

Simpson’s paradox occurs when a relationship that holds within every subgroup reverses when the subgroups are aggregated. It arises when a lurking variable is distributed unevenly across the groups.

Worked example. A university compares admission rates of two departments. Within each department women are admitted at a higher rate than men, yet aggregated over both departments men have the higher overall rate. The reversal happens because women applied mostly to the more competitive department, which admits fewer applicants of either sex. Diagnosing the paradox means computing the rates both within subgroups and in aggregate and identifying the variable whose uneven distribution causes the reversal.

5 Data visualization

Choosing the right chart depends on the data type:

- **Bar chart:** compare a numeric value across categories.
- **Histogram:** show the distribution of one numeric variable by binning it.
- **Scatter plot:** show the relationship between two numeric variables.
- **Box plot:** compare distributions and show spread and outliers across groups.
- **Heatmap:** show a matrix of values by colour, for example a correlation matrix.

5.1 Misleading visualizations

Common ways a chart misleads:

- A bar-chart axis that does not start at zero exaggerates differences.
- A truncated or inconsistent axis scale distorts trends.
- Inconsistent bin widths in a histogram misrepresent the distribution.
- Encoding a one-dimensional quantity by area or volume exaggerates ratios.
- Omitting data, cherry-picking a time window, or using a dual axis to suggest a relationship.

6 Exploratory data analysis

Exploratory data analysis (EDA) is a workflow for understanding a dataset before formal modelling:

1. **State a hypothesis** or question about the data.
2. **Compute summary statistics** — central tendency and dispersion for each variable.
3. **Visualize distributions** — histograms and box plots for single variables, scatter plots for pairs.
4. **Detect outliers** — for example with the $1.5 \times \text{IQR}$ rule or by inspecting box plots.
5. **Examine missing data** — count missing values per variable and look for patterns (whether values are missing at random or systematically).

The shape of a distribution (symmetric, skewed, unimodal, multimodal) guides which central-tendency and dispersion measures are appropriate.

7 Bridge to Foundations: scale and transformations

Summary statistics depend on the scale and units of the data, and monotone transformations change them in predictable ways.

- **Linear rescaling:** if $y = ax + b$, then $\bar{y} = a\bar{x} + b$, $s_y = |a|s_x$, and $s_y^2 = a^2 s_x^2$. Changing units (for example °C to °F) is such a rescaling.
- **Logarithmic transformation:** $y = \log x$ compresses a right-skewed distribution toward symmetry and turns multiplicative growth into additive steps. It is appropriate for positive data spanning several orders of magnitude.
- **z-score (standardization):**

$$z_i = \frac{x_i - \bar{x}}{s_x}.$$

The standardized data have mean 0 and standard deviation 1, which makes variables on different scales directly comparable.

A **growth rate** is the relative change $\frac{x_{t+1}-x_t}{x_t}$; on a logarithmic scale, constant growth appears as a straight line, so log transforms are used to read and compare growth.

Common pitfall Pearson correlation is invariant under increasing linear rescaling of each variable, but a non-linear transformation such as log changes the correlation, because it changes the linear structure of the relationship.

8 Bridge to Linear Algebra I

Centre each variable by subtracting its mean: let $\tilde{x}_i = x_i - \bar{x}$ and $\tilde{y}_i = y_i - \bar{y}$, and collect them into vectors $\tilde{\mathbf{x}}, \tilde{\mathbf{y}} \in \mathbb{R}^n$. Then

$$\text{cov}(x, y) = \frac{1}{n}(\tilde{\mathbf{x}} \cdot \tilde{\mathbf{y}}), \quad s_x = \frac{1}{\sqrt{n}}\|\tilde{\mathbf{x}}\|, \quad r = \frac{\tilde{\mathbf{x}} \cdot \tilde{\mathbf{y}}}{\|\tilde{\mathbf{x}}\|\|\tilde{\mathbf{y}}\|} = \cos \theta,$$

where θ is the angle between the centred vectors. So covariance is the dot product of the centred data — a measure of geometric co-movement — and the correlation is the cosine of the angle between them: $r = \pm 1$ means the vectors are parallel, $r = 0$ means they are orthogonal.

Fitting a linear trend $y \approx ax + b$ to a scatter plot corresponds to **projecting** the centred response vector onto the direction of the centred predictor: the fitted values are the projection and the residuals are the orthogonal component, connecting the linear trend to the projection concept from Linear Algebra I.

9 Python

The analysis is carried out reproducibly with pandas for data manipulation and matplotlib/seaborn for visualization.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("data.csv")

# Cleaning
df = df.dropna(subset=["x", "y"]) # handle missing data

# Summary statistics
df.describe() # mean, std, quartiles
df["x"].median()
df[["x", "y"]].cov() # covariance
df[["x", "y"]].corr() # Pearson correlation

# Transformations
import numpy as np
df["log_x"] = np.log(df["x"])
df["z_x"] = (df["x"] - df["x"].mean()) / df["x"].std()
```

```
# Visualization
sns.histplot(df["x"])
sns.boxplot(x=df["x"])
sns.scatterplot(data=df, x="x", y="y")
sns.heatmap(df.corr(), annot=True)
plt.show()
```

A reproducible analysis records the cleaning steps, the visualizations, and a written interpretation that states at least one limitation of the data or method.

Exercises

1. Compute the mean, median, and mode of a dataset and justify which measure of central tendency is appropriate given its distribution shape and outliers.
2. Compute the variance, standard deviation, and IQR of a dataset and interpret each in the context of the data-generating process.
3. Draw a box plot from a five-number summary and mark the outliers using the $1.5 \times$ IQR rule.
4. Compute the covariance and Pearson correlation of two variables, and interpret the sign and magnitude of r .
5. Give an example of two correlated variables that are not causally related, and identify the common cause.
6. For a given study, identify whether sampling, selection, survivorship, or confirmation bias is present and explain how it distorts the conclusion.
7. Construct or analyse a dataset exhibiting Simpson's paradox and identify the variable whose uneven distribution causes the reversal.
8. For given data types, choose an appropriate chart, and identify a misleading feature in a given chart.
9. Conduct an EDA workflow on a dataset: state a hypothesis, compute summary statistics, visualize distributions, and detect outliers.
10. Apply a logarithmic transformation and a z-score transformation to a dataset and explain how each changes the summary statistics.
11. Centre two variables, compute the correlation as the cosine of the angle between the centred vectors, and explain the projection interpretation of a linear trend.
12. Produce a reproducible pandas + matplotlib/seaborn analysis with data cleaning, visualization, and a written interpretation stating one limitation.